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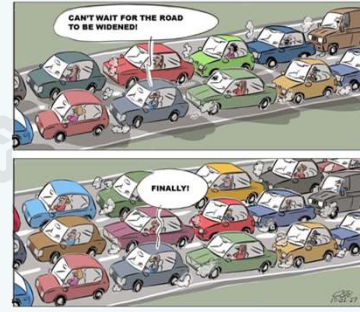
Traffic Shaping in ROSv7

BelajarMikroTik x APJII Jatim ROADSHOW
Herry Darmawan

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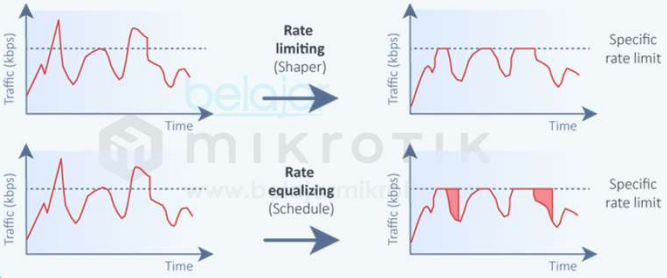
Traffic Shaping



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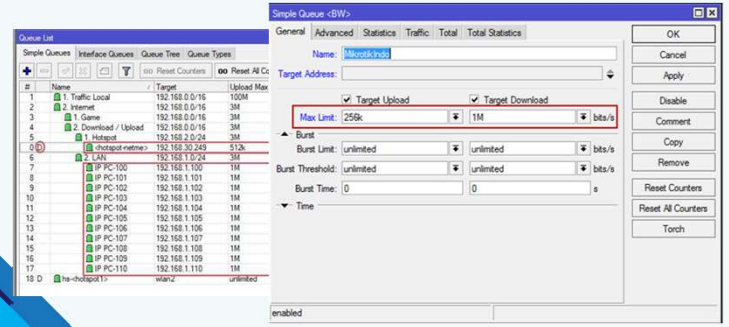
Rate Limiting



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Rate Limit in ROS



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Queue Type

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Queue Type

- Define how the queue is managed
- Mechanism to select which packet goes next

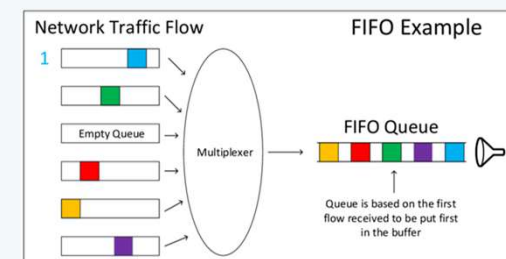
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Queue Type

- | | |
|-------------------------------|------------|
| • ROSv6 | • ROSv7 + |
| - FIFO (pfifo, bfifo, mqfifo) | - CoDel |
| - RED | - FQ-CoDel |
| - SFQ | - CAKE |
| - PCQ | |

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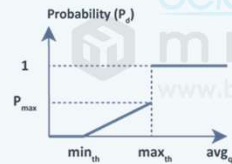
Queue Type - FIFO



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Queue Type - RED

- Random Early Drop
- Ideal for : Serial Connection

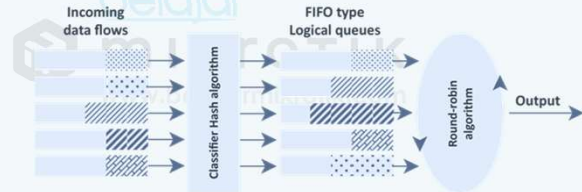


$avg_q \leq min_{th}$ - no dropped packets
 $avg_q \geq max_{th}$ - dropped all packets
 $min_{th} \leq avg_q \leq max_{th}$ - packets are dropped with probability P_d

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Queue Type - SFQ

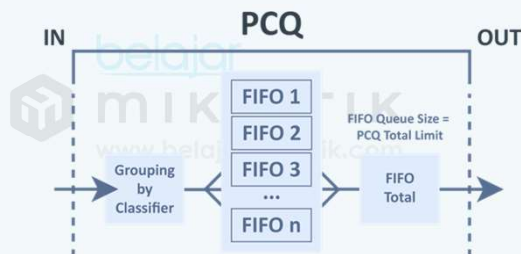
- Stochastic Fairness Queue
- Ideal for : HOTSPOT and wireless



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Queue Type - PCQ

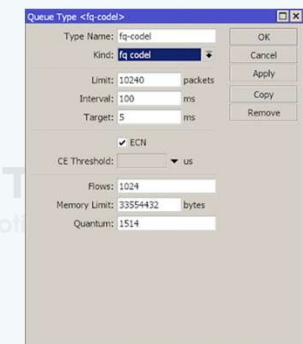
- Per-Connection Queue



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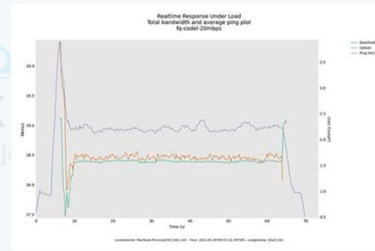
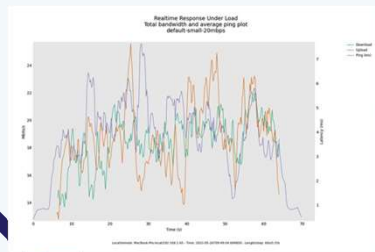
Queue Type - CoDel

- Controlled Delay Active Queue Management
- Fair Queueing (FQ) with Controlled Delay (CoDel)



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Queue Type - CoDel



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Queue Type - CAKE



- Common Application Keep Enhanced
- qdisc for Linux Kernel COBALT (CoDel+BLUE)
- Automatically enhanced delay and rate based on common application

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Queue Simple x Queue Tree



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Similarity



VS

New Simple Queue

General | Advanced | Statistics | Traffic | Total | Total Statistics

Name: queue1

Target: 0.0.0.0/0

Dist.:

Max Limit: 0

Burst Limit: 0

Burst Threshold: 0

Burst Time: 0

Time: 00:00:00

Days: ☐ sun ☐

Limit At: 0

Priority: 8

Bucket Size: 0.100

Queue Type: default-small

Parent: none

New Queue

General | Statistics

Name: queue1

Parent: global

Packet Marks:

Queue Type: default-small

Bucket Size: 0.100

Limit At:

Max Limit:

Burst Limit:

Burst Threshold:

Burst Time:

enabled

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Simple Queue

- Based on Target (IP or Address)
- Time
- Dst
- Automatic

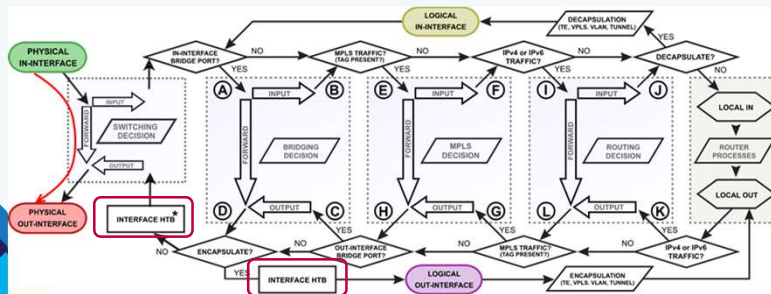
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Queue Tree

- Based on Interface
- Depend on Packet Mark
- Manual

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Queue in the Packet Flow



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Simple Queue vs QTree

- Simple Queue

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Simple Queue vs QTree



. Queue Tree

Name	Parent	Packet Marks
Client 1-Do...	ether2	Packet-Client1
Client 1-Up...	ether1	Packet-Client1

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Simple Queue vs QTree



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Packet Flow



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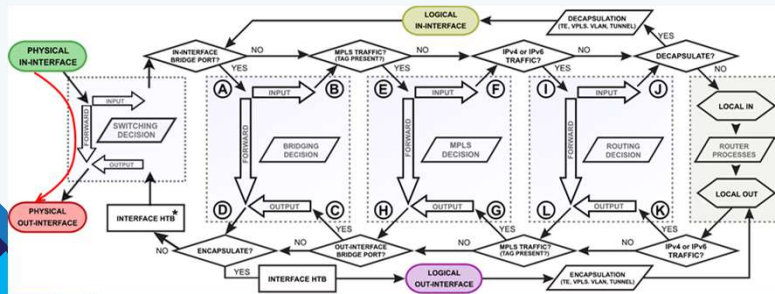
What is Packet Flow



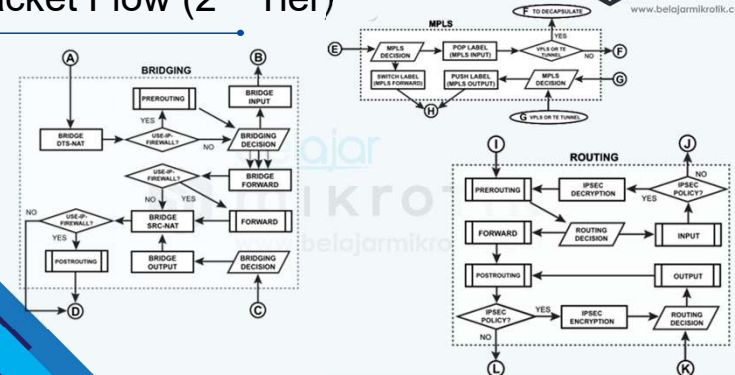
- . The flowchart that show where packet goes in the router (from start to end)
- . Knowing where the packet goes can define where we put our rules to identify packet

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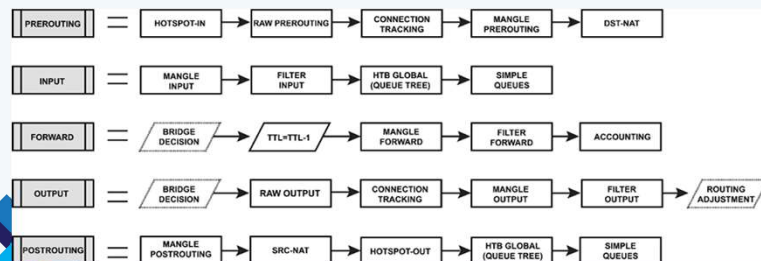
Packet Flow (1st Tier)



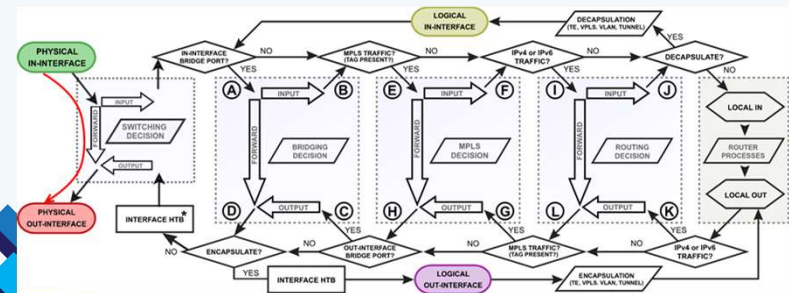
Packet Flow (2nd Tier)



Packet Flow (3rd Tier)



Packet Flow (1st Tier)



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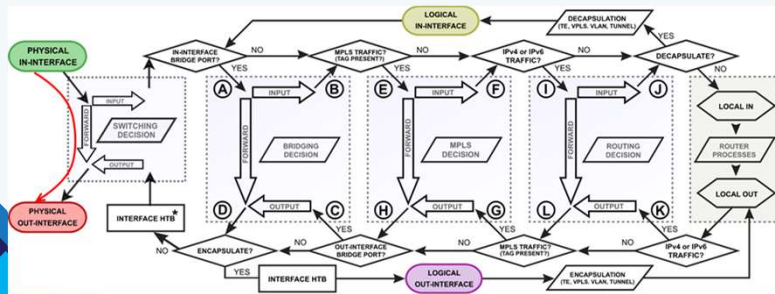
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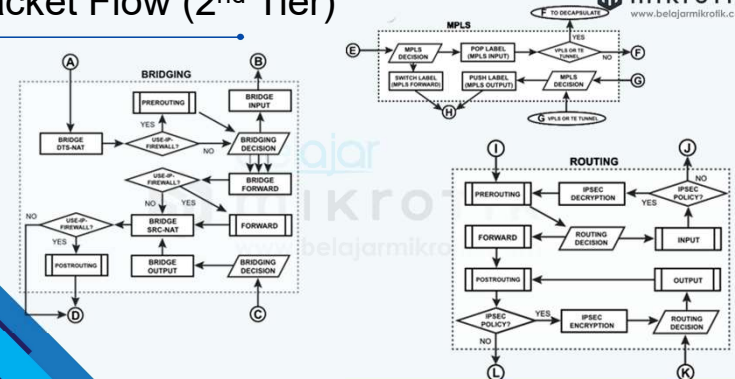
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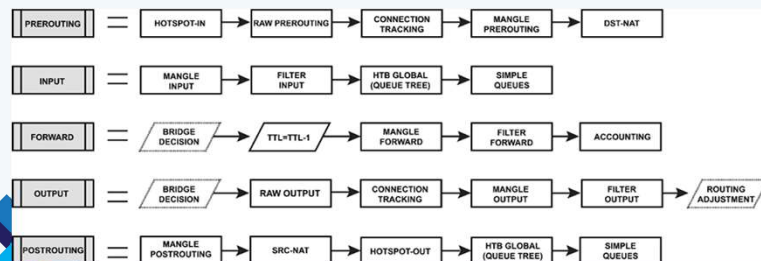
Packet Flow (1st Tier)



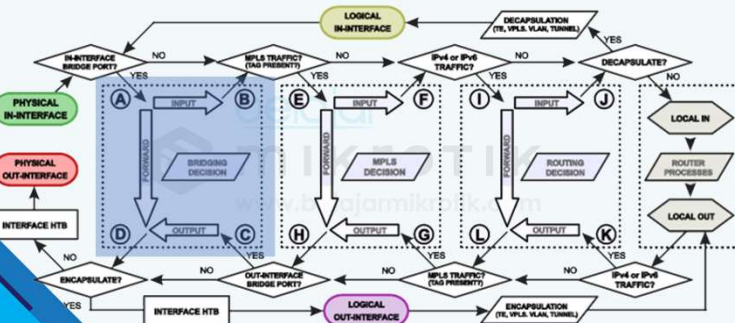
Packet Flow (2nd Tier)



Packet Flow (3rd Tier)

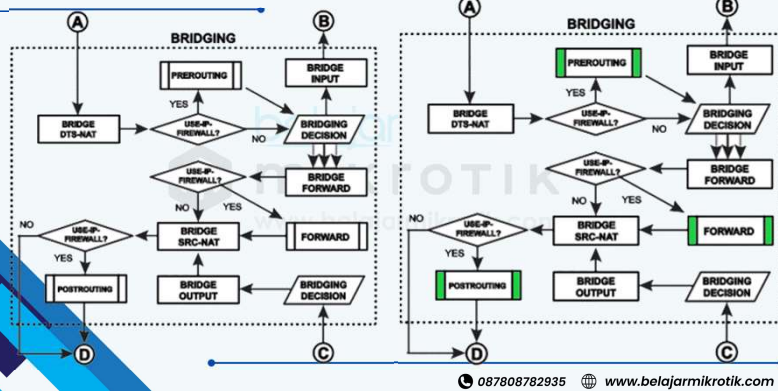


SECTION #1 – Bridge (LAYER2)



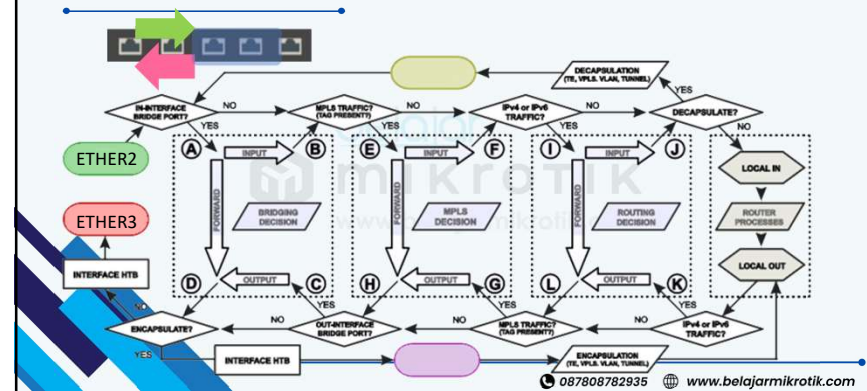
SECTION #1 – Bridge (LAYER2)

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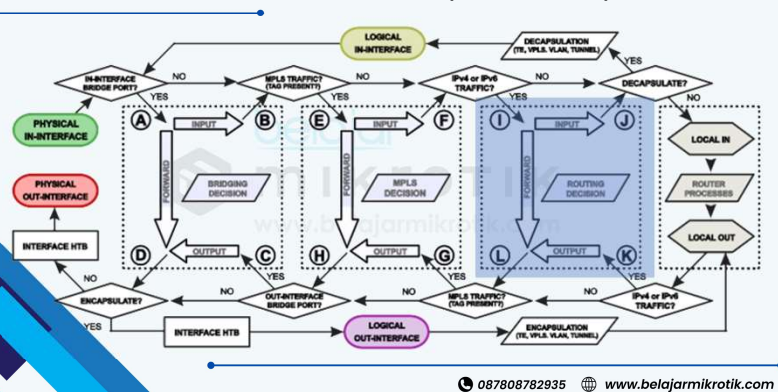
SECTION #1 – Bridge (LAYER2)

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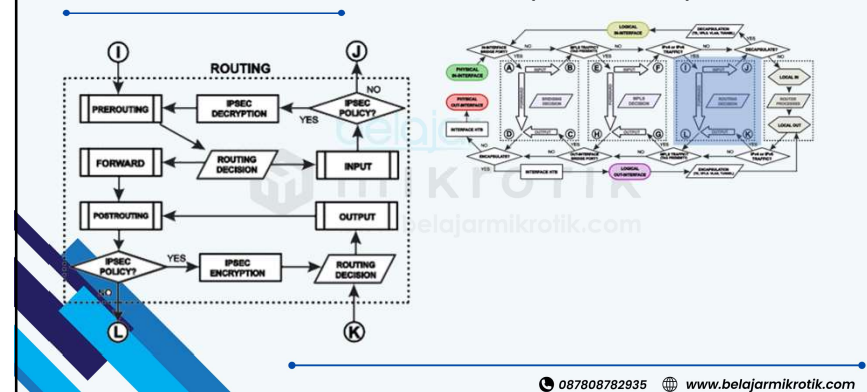
SECTION #2 – ROUTING (LAYER3)

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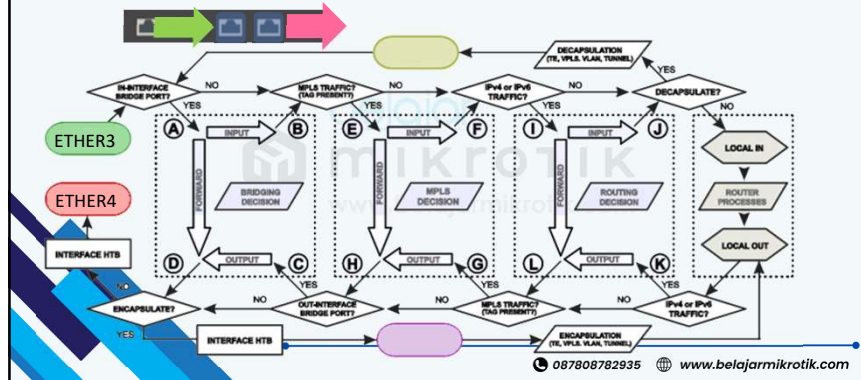
SECTION #2 – ROUTING (LAYER3)

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SECTION #2 – ROUTING (LAYER3)

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Parent/Child Queue with Priority

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Parent/Child Queue

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- Based on Hierarchical Token Bucket
- Applied originally in Queue Tree but nowadays can be used in Simple Queue as well

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HTB

New Queue

General

Name: queue1

Parent: global

Packet Marks: default-small

Queue Type: default-small

Priority: 8

Bucket Size: 0.100

Limit At: bits/s

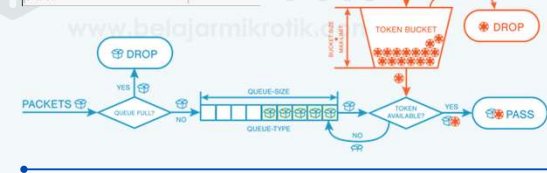
Max Limit: bits/s

Burst Limit: bits/s

Burst Threshold: bits/s

Burst Time: s

enabled



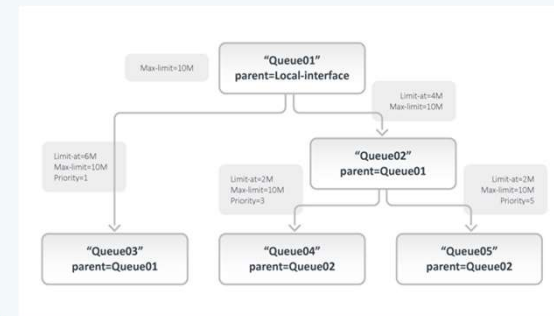
Principle of Parent/Child Queue



- Traffic will fill out the LIMIT-AT of child queue
- Traffic then will fill out the LIMIT-AT of parent
- If there is any extra bandwidth, it will be given to the highest priority of the child queue
- If any extra bandwidth left, it will be given to the next queue with higher priority

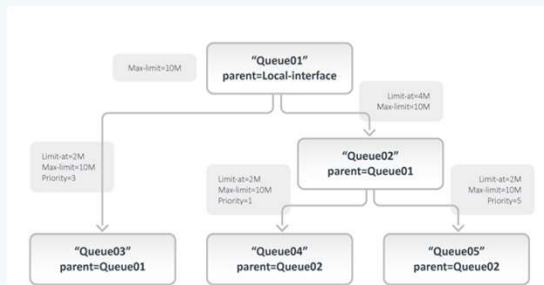
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Parent/Child – Case Study 1



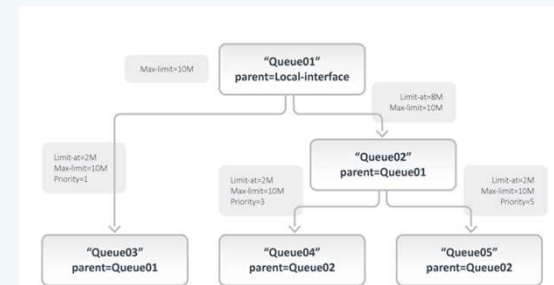
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Parent/Child – Case Study 2



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Parent/Child – Case Study 3



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```

graph TD
    Queue01["Queue01  
parent=Local-interface"]
    Queue03["Queue03  
parent=Queue01"]
    Queue02["Queue02  
parent=Queue01"]
    Queue04["Queue04  
parent=Queue02"]
    Queue05["Queue05  
parent=Queue02"]

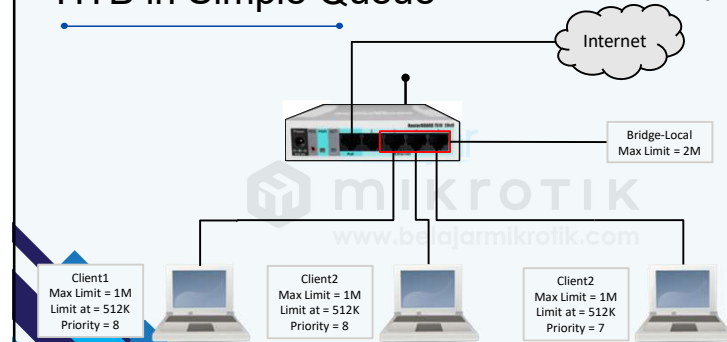
    Queue01 --> Queue03
    Queue01 --> Queue02
    Queue01 --> Queue05
    Queue02 --> Queue04
    Queue02 --> Queue05
  
```

Diagram illustrating the configuration of queues and their parents:

- Queue01** (parent=Local-interface)
 - Limit-at=6M, Max-limit=10M, Priority=1
 - Queue03 (parent=Queue01)
 - Queue02 (parent=Queue01)
 - Limit-at=2M, Max-limit=10M, Priority=3
 - Queue04 (parent=Queue02)
 - Queue05 (parent=Queue02)
- Queue02** (parent=Queue01)
 - Limit-at=4M, Max-limit=10M
 - Queue04 (parent=Queue02)
 - Queue05 (parent=Queue02)
- Queue03** (parent=Queue01)
- Queue04** (parent=Queue02)
- Queue05** (parent=Queue02)

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Queue List

Simple Queues

Interface Queues

Queue Tree

Queue Types

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- Client 1?
- Client 2?
- Client 3?

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ISP Traffic Shaping

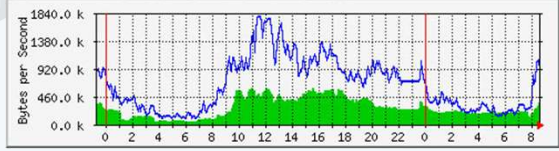
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What to Consider???

- Day/Night Traffic
- Upload/Download Traffic
- Sharing Traffic (1:8, or 1:4, or 1:5)



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Q&A

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